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1 AN AUTONOMOUS AI COMPANION ROBOT

Robots, though initially used for purposes like manufacturing in factories, are slowly finding their way more often in workplaces and homes for day-to-day work. After realising how troublesome it can be for older people to perform basic tasks with their own hands, Yossi Wolf, co-founder of Roboteam, designed the autonomous personal artificial intelligence (AI) assistant robot Temi that offers high performance with minimal power consumption.

Weighing 12kg, the robot (100cm × 35cm × 45cm) has a 25.65cm HD LCD screen with fingerprint-resistant coating and autonomous face tracking screen tilt (15° ~ +55°). It uses Robox autonomous navigation system for 3D mapping, user detection and tracking, obstacle avoidance, path planning, and so on. This enables mobility and stability over any kind of surfaces such as hard floors, rugs, tiles, and room transitions; its adaptive motor control allows adjustment of torque accordingly. A system of sensors (like 360 degrees LIDAR, depth cameras, RGB camera, proximity sensors) is present to sense an obstacle and move around it as well as algorithms to stop if it cannot be avoided, such as a staircase. The system supports both Bluetooth and Wi-Fi connectivity.

The built-in AI feature lets users utilise Amazon's 'Alexa' or 'Hey temi' as a voice assistant within their applications. It has auto speech recognition (ASR) and far-field voice technology, natural language processing (NLP), speech-to-text, and text-to-speech (TTS) engines to respond to the user.

Multiple cameras (13MP high-resolution camera, 13MP wide camera for remote navigation, and a time-of-flight (TOF) depth camera) are available for capturing images.

Other features include four omnidirectional digital mics with acoustic echo cancellation and environment noise reduction; lithium-ion battery with up to eight hours of operation per charge and a docking station for recharging; 20W audio with one sub-woofer, two midrange speakers, two tweeters, passive radiators, and high fidelity equaliser; custom tires with a zero-turn radius and a speed up to one metre per second, among others.

The temi app (available for free on iOS and Android) enables users to control the bot from anywhere in the world and virtually move through their homes to make sure everything is fine in their absence. The app supports English, Japanese, Korean, simplified Chinese, along with the inclusion of Thai and traditional Chinese in the latest version. The software gets updated over-the-air whenever needed. The app allows users to interact and share photos and videos instantly with people who do not have the assistant.

The bot can identify and follow when asked, save preset locations, and navigate around autonomously while simultaneously controlling all smart home devices and connecting users to other people through its dynamic audio and video feature and web services. User data is fully encrypted with the latest standards when sent over the Internet or saved



in the cloud storage platform.

Besides being a personal robot for home use, it serves as an engagement solution for retail, hospitality, enterprise, education, healthcare, entertainment, and so on. Recently, the US-based startup announced a global partnership with Connected Living community in response to the Covid-19 pandemic.

—Ayushee Sharma